Importing necessary libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as  plt
import seaborn as sns
import plotly.express as px
from plotly.offline import init_notebook_mode, iplot
init_notebook_mode(connected=True)
```

Loading Dataset

In [2]:
```
df = pd.read_csv("resolved_tickets.csv")
```

Exploratory Data Analysis(EDA)

In [3]:
```
df.head()
```

Out[3]:

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	D. Pur
0	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	202
1	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	202
2	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	202
3	11	Joseph Moreno	mbrown@example.org	48	Male	Nintendo Switch	202
4	12	Brandon Arnold	davisjohn@example.net	51	Male	Microsoft Xbox Controller	202

5 rows × 24 columns

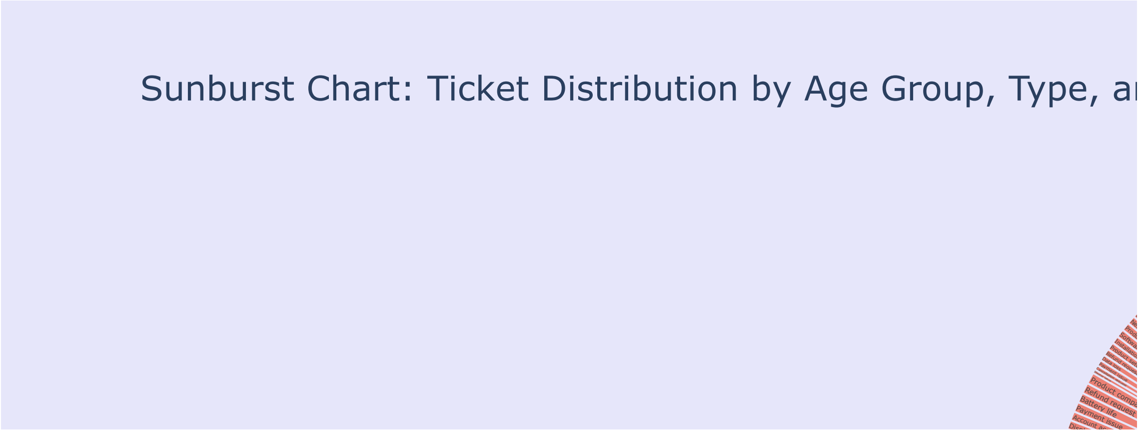
In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2769 entries, 0 to 2768
Data columns (total 24 columns):
 #   Column                          Non-Null Count  Dtype  
---  ------                          --------------  -----  
 0   Ticket ID                       2769 non-null   int64  
 1   Customer Name                   2769 non-null   object 
 2   Customer Email                  2769 non-null   object 
 3   Customer Age                    2769 non-null   int64  
 4   Customer Gender                 2769 non-null   object 
 5   Product Purchased               2769 non-null   object 
 6   Date of Purchase                2769 non-null   object 
 7   Ticket Type                     2769 non-null   object 
 8   Ticket Subject                  2769 non-null   object 
 9   Ticket Description              2769 non-null   object 
 10  Ticket Status                   2769 non-null   object 
 11  Resolution                      2769 non-null   object 
 12  Ticket Priority                 2769 non-null   object 
 13  Ticket Channel                  2769 non-null   object 
 14  First Response Time             2769 non-null   object 
 15  Time to Resolution              2769 non-null   object 
 16  Customer Satisfaction Rating    2769 non-null   float64
 17  Age Group                       2769 non-null   object 
 18  Is Resolved                     2769 non-null   bool   
 19  Cleaned_Description             2769 non-null   object 
 20  Response Delay (in hours)       2769 non-null   float64
 21  Resolution Time (in hours)      2769 non-null   float64
 22  Customer_Response_Pending_Days  2769 non-null   float64
 23  Auto_Close                      2769 non-null   bool   
dtypes: bool(2), float64(4), int64(2), object(16)
memory usage: 481.5+ KB
```

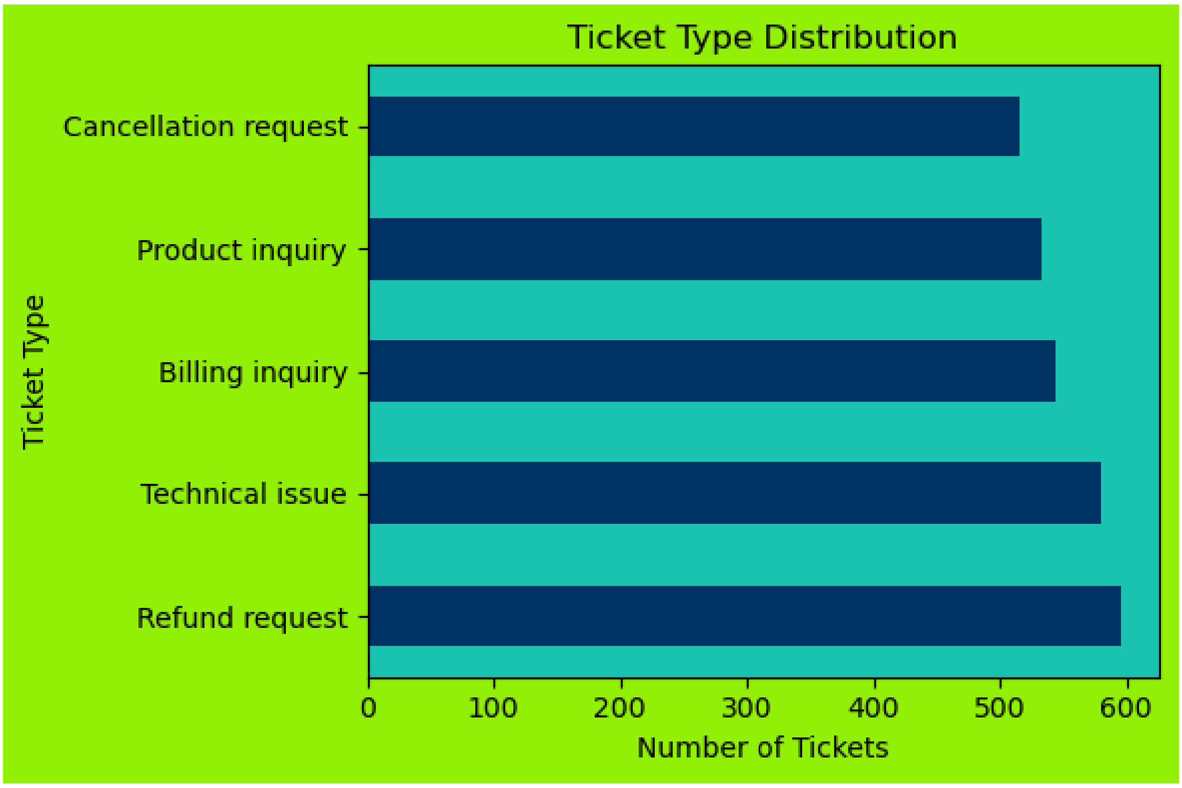
In [5]:
```
df['Customer Satisfaction Rating'].value_counts()
```

Out[5]:
```
Customer Satisfaction Rating
3.0    580
1.0    553
2.0    549
5.0    544
4.0    543
Name: count, dtype: int64
```

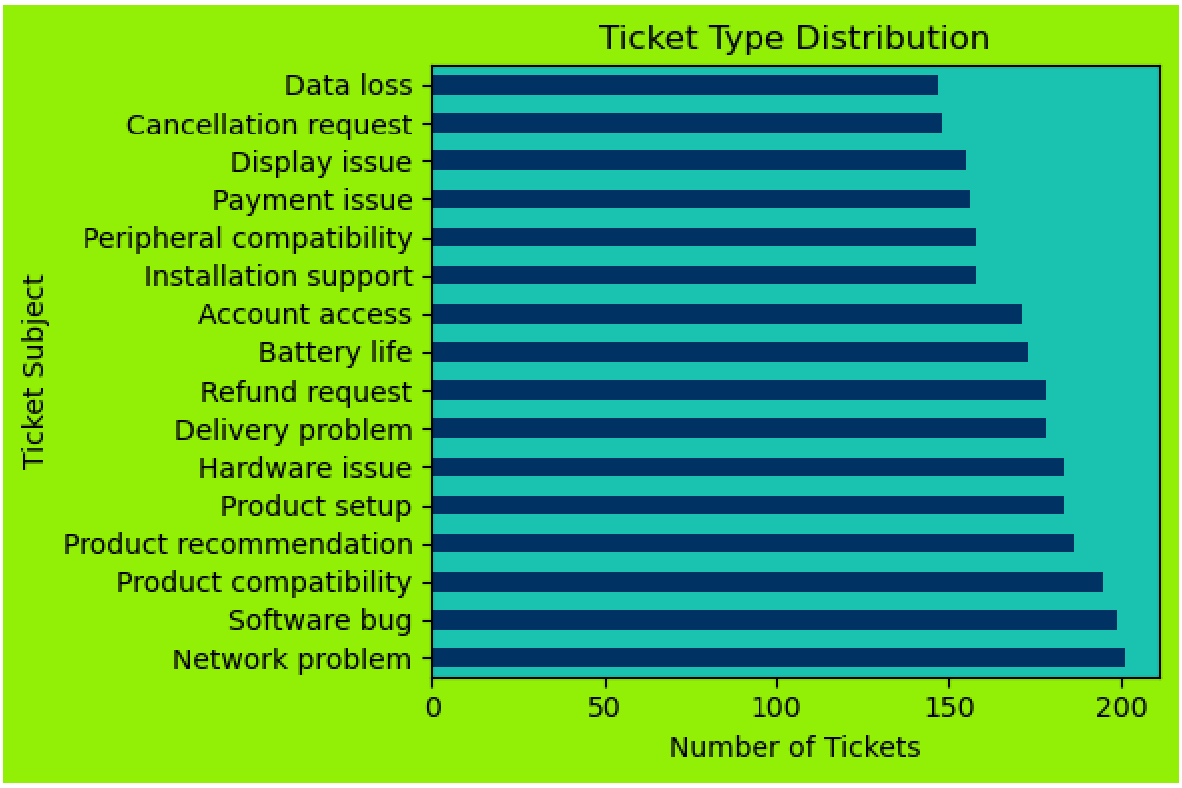
In [6]:
```
sunburst_data = df.groupby(['Age Group', 'Ticket Type', 'Ticket Subject']).size(

fig = px.sunburst(
    sunburst_data,
    path=['Age Group', 'Ticket Type', 'Ticket Subject'],
    values='Count',
    title="Sunburst Chart: Ticket Distribution by Age Group, Type, and Subject"
)

fig.update_layout(
    paper_bgcolor="#E6E6FA",

)
iplot(fig)
```



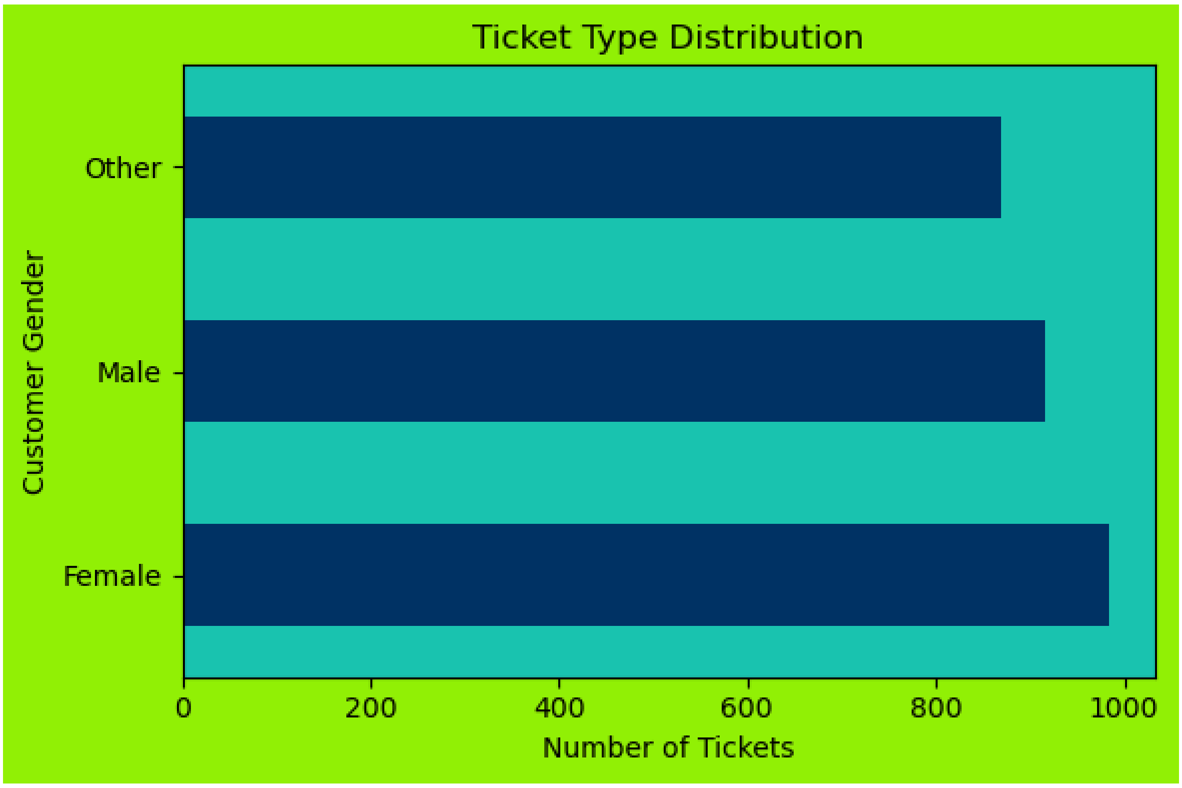
In [7]:
```
# Counting the number of each ticket type
ticket_counts = df['Ticket Type'].value_counts()

# Plotting  bar chart
plt.figure(figsize=(6,4),facecolor= "#94F008")
ax= ticket_counts.plot(kind='barh',color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Type')
plt.tight_layout()
plt.show()
```



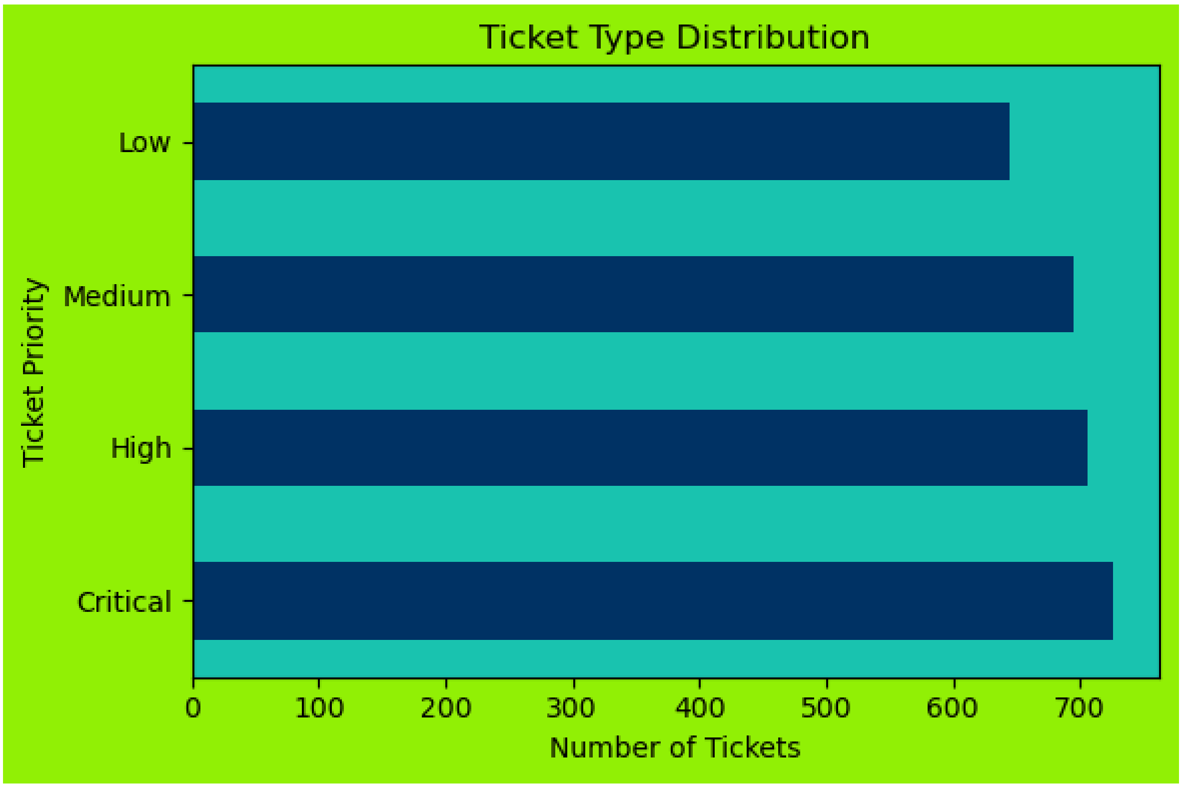
In [8]:
```python
# Counting the number of each ticket subject
ticket_counts = df['Ticket Subject'].value_counts()

# Plotting  bar chart
plt.figure(figsize=(6,4),facecolor= "#94F008")
ax = ticket_counts.plot(kind='barh',color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Subject')
plt.tight_layout()
plt.show()
```



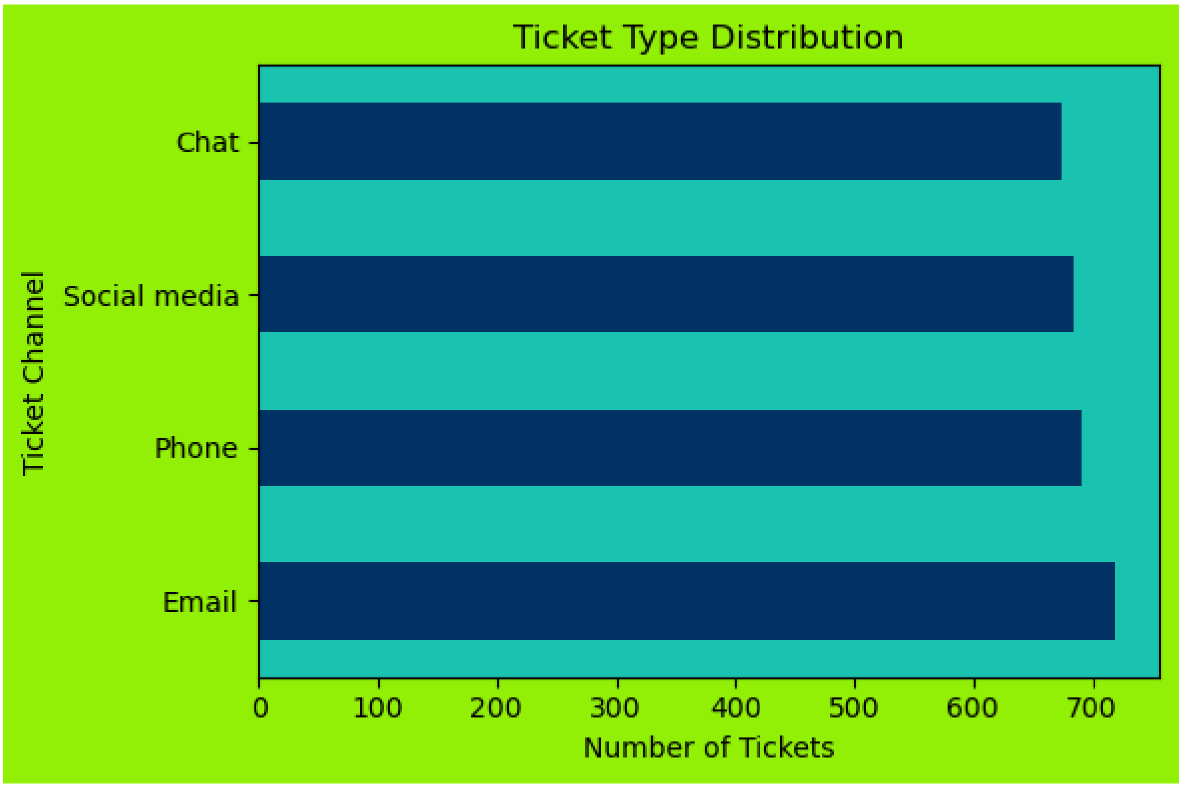
In [9]:
```
# Counting Gender
ticket_counts = df['Customer Gender'].value_counts()

# Plotting bar chart
plt.figure(figsize=(6,4),facecolor= "#94F008")
ax = ticket_counts.plot(kind='barh',color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Customer Gender')
plt.tight_layout()
plt.show()
```



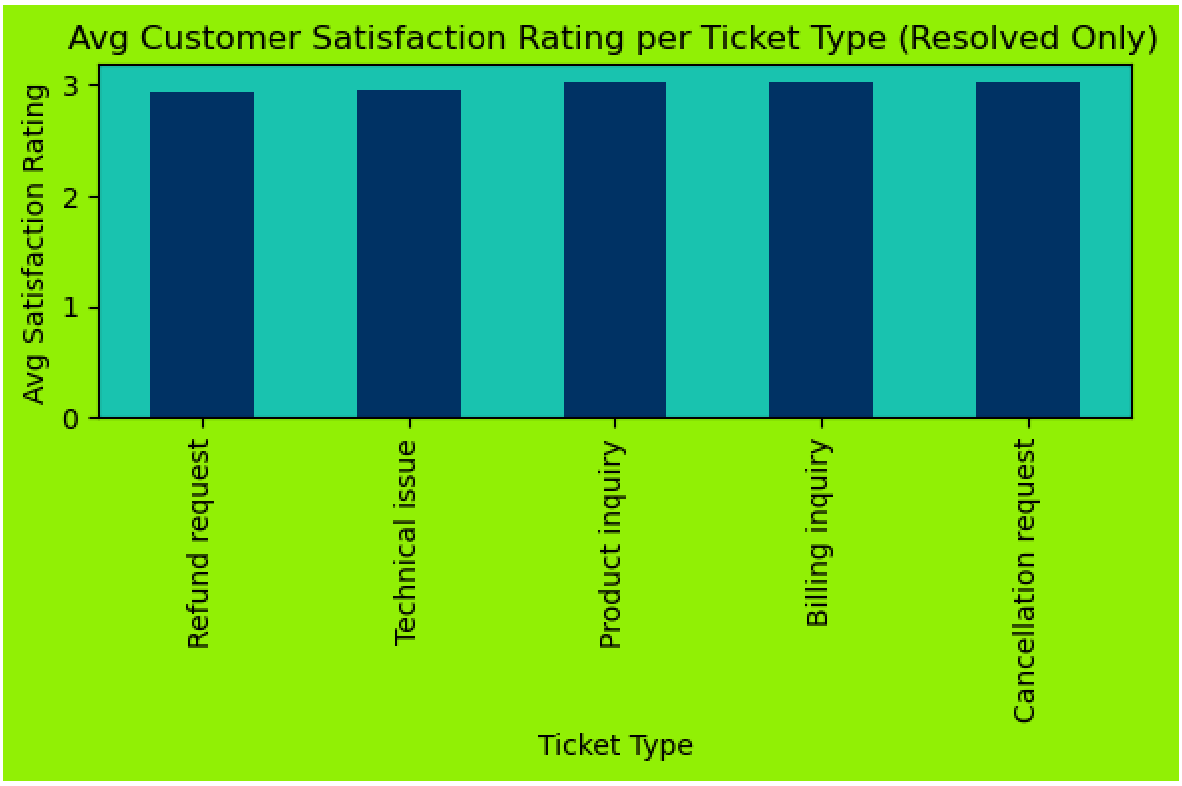
In [10]:
```
# Count the number of each ticket priority
ticket_counts = df['Ticket Priority'].value_counts()

# Plotting bar chart
plt.figure(figsize=(6,4),facecolor= "#94F008")
ax = ticket_counts.plot(kind='barh',color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Priority')
plt.tight_layout()
plt.show()
```



In [11]:
```
# Count the number of each ticket channel
ticket_counts = df['Ticket Channel'].value_counts()

# Plotting bar chart
plt.figure(figsize=(6,4),facecolor= "#94F008")
ax = ticket_counts.plot(kind='barh',color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Channel')
plt.tight_layout()
plt.show()
```



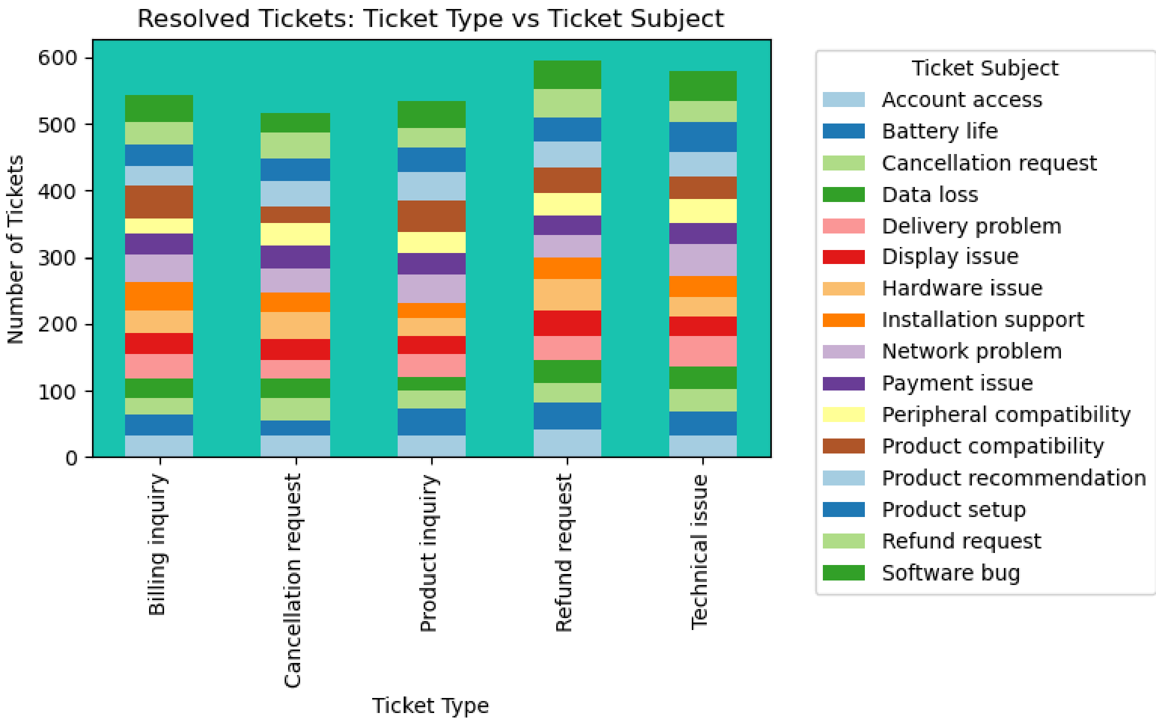
In [12]:
```
resolved_df = df[df['Is Resolved'] == True]

# Grouping by Ticket Type and calculating average satisfaction rating
avg_rating = resolved_df.groupby('Ticket Type')['Customer Satisfaction Rating'].

# Plotting bar chart
plt.figure(figsize=(6, 4), facecolor="#94F008")
ax = avg_rating.plot(kind='bar', color="#003366")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Avg Customer Satisfaction Rating per Ticket Type (Resolved Only)')
plt.ylabel('Avg Satisfaction Rating')
plt.xlabel('Ticket Type')
plt.tight_layout()
plt.show()
```



In [13]:

```
# Filtered only resolved tickets
resolved_df = df[df['Is Resolved'] == True]

# Grouped by Ticket Type and Ticket Subject  count
grouped = resolved_df.groupby(['Ticket Type', 'Ticket Subject']).size().unstack(

# Plotting stacked bar chart
plt.figure(figsize=(8, 5), facecolor="#94F008")
ax = grouped.plot(kind='bar', stacked=True, figsize=(8, 5), color=plt.cm.Paired.
ax.set_facecolor("#1CC4AF")

#plot
plt.legend(title="Ticket Subject", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.title('Resolved Tickets: Ticket Type vs Ticket Subject')
plt.xlabel('Ticket Type')
plt.ylabel('Number of Tickets')
plt.tight_layout()
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```



In [14]:

```
# Filtered only resolved tickets with positive resolution time
filtered_df = df[(df['Is Resolved'] == True) & (df['Resolution Time (in hours)']

# Grouped by Customer Satisfaction Rating and calculated average resolution time
avg_resolution = filtered_df.groupby('Customer Satisfaction Rating')['Resolution

# Plot
plt.figure(figsize=(6, 4), facecolor="#94F008")
ax = avg_resolution.plot(kind='bar', color="#003366")
ax.set_facecolor("#1CC4AF")

plt.title('Avg Resolution Time by Customer Satisfaction Rating')
plt.xlabel('Customer Satisfaction Rating')
plt.ylabel('Avg Resolution Time (hours)')
plt.tight_layout()
plt.show()
```

Summary of Resolved Ticket Data (Elderly Users Focus)

What we found:
Elderly users experience the most issues compared to other age groups. Most of their problems are related to **technical issues** (like software bugs, setup, compatibility) and **refund requests** (payment and refund processing).

What this means:
Elderly users may face difficulties with complex processes, highlighting the need for improved support tailored to their needs.

What we can do:

- We can implement a **chatbot** that guides users through common issues like technical problems, product enquiry etc... This will streamline support and be especially helpful for elderly users who may struggle with complex navigation or waiting for responses.
- We can enhance the **Help Section** by adding a clearly visible, easy-to-understand FAQ page that addresses frequently raised concerns.
- Both the chatbot and the improved help section can empower users to find answers on their own, reducing the number of tickets raised and allowing the support team to focus on more critical cases.

Customer Ratings Insight:
Elderly users tend to give lower satisfaction ratings due to delays or unresolved

issues. Providing quick access to solutions will help increase their satisfaction and reduce frustration.

In [ ]: